

Additional file 1: Supplementary tables

No Single Architecture Wins: Ensembles and Conventional Machine Learning
Match Pretrained Molecular Models Across 45 Property-Prediction Benchmarks

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This file contains the four supplementary tables referenced in the main text. All are generated directly from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`, and carry the same values as the machine-readable `manuscript_assets/tables/*.csv` files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

dataset	suite	task - kind	family	model	analysis_metric	analysis_metric_value	cv_selected_model	cv_selected_family	cv_selected_gap_to_best
tdc_ames	TDC	classification	Conventional ML	XGBoost	test_roc_auc	0.875	Extra trees	Conventional ML	0.777
tdc_bbb_martins	TDC	classification	Conventional ML	Random forest	test_roc_auc	0.932	Tabular MLP	Conventional ML	5.108
tdc_bioavailability_ma	TDC	classification	Conventional ML	CatBoost	test_roc_auc	0.777	Logistic Regression	Conventional ML	7.943
tdc_carcinogens_lagunin	TDC	classification	Conventional ML	AdaBoost	test_roc_auc	0.863	Logistic Regression	Conventional ML	3.747
tdc_clintox	TDC	classification	Conventional ML	CatBoost	test_roc_auc	0.949	Logistic Regression	Conventional ML	3.552
tdc_cyp1a2_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.970	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	2.383
tdc_cyp2c19_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.921	XGBoost	Conventional ML	8.809
tdc_cyp2c9_substrate_carbonmangels	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.482	SVC	Conventional ML	37.538
tdc_cyp2c9_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.810	AdaBoost	Conventional ML	12.949
tdc_cyp2d6_substrate_carbonmangels	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_auprc	0.673	AdaBoost	Conventional ML	3.067
tdc_cyp2d6_veith	TDC	classification	Conventional ML	XGBoost	test_auprc	0.722	AdaBoost	Conventional ML	14.842
tdc_cyp3a4_substrate_carbonmangels	TDC	classification	Conventional ML	LogisticRegression	test_auprc	0.717	Tabular MLP	Conventional ML	5.325
tdc_cyp3a4_veith	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOF Stacking (RidgeCV, 5-fold))	test_auprc	0.890	AdaBoost	Conventional ML	10.365

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dataset	suite	task - kind	family	model	analysis - metric	analysis_metric_value	cv_se- lected - model	cv - selected - family	cv_selected_gap_to_best
tdc_dili	TDC	classification	Ensemble (stacking / averaging)	Ensemble (OOFCV, 5-fold))	test_roc_auc	0.915	Logistic Regression	Conventional ML	6.036
tdc_herg	TDC	classification	Conventional ML	AdaBoost	test_roc_auc	0.861	Tabular MLP	Conventional ML	15.894
tdc_herg_karim	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.902	SVC	Conventional ML	9.739
tdc_hia_hou	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_roc_auc	0.990	Logistic Regression	Conventional ML	0.270
tdc_pampa_ncats	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.806	Tabular MLP	Conventional ML	5.601
tdc_pgp_broccatelli	TDC	classification	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_roc_auc	0.929	Tabular MLP	Conventional ML	2.457
tdc_skin_reaction	TDC	classification	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_roc_auc	0.769	Logistic Regression	Conventional ML	4.572
tdc_tox21	TDC	classification	Conventional ML	XGBoost	test_roc_auc	0.812	SVC	Conventional ML	2.861
tdc_toxcast	TDC	classification	Conventional ML	CatBoost	test_roc_auc	0.791	CatBoost	Conventional ML	0.000
chemml_cep_homo	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_rmse	0.086	Elastic NetCV	Conventional ML	28.254
chemml_organic_density	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_rmse	0.005	Elastic NetCV	Conventional ML	26.007
chemml_xyz_polarizability	ChemML	regression	TabPFN (tabular foundation)	TabPFNRegressor	test_rmse	0.007	TabPFNRegressor	TabPFN (tabular foundation)	0.000
esol_delaney	MolecularNet	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_rmse	0.592	Elastic NetCV	Conventional ML	14.805
freesolv_sampl	MolecularNet	regression	Chemprop v2 GNN	Chemprop v2 (AttentiveFP, ensemble=1)	test_rmse	1.080	Elastic NetCV	Conventional ML	15.792
lipophilicity	MolecularNet	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_rmse	0.582	Elastic NetCV	Conventional ML	13.041

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_ metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
poduam_pod_ nc_std	PODUAM	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.699	XGBoost	Conventional ML	5.154
poduam_pod_ rd_std	PODUAM	regression	Conventional ML	XGBoost	test_ rmse	0.551	Elastic NetCV	Conventional ML	29.477
polaris_adme_ fang_hppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.449	Elastic NetCV	Conventional ML	16.167
polaris_adme_ fang_perm_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.399	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	18.507
polaris_adme_ fang_rclint_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.512	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	12.570
polaris_adme_ fang_rppb_1	Polaris	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	0.494	TabPFNReg	TabPFN (tabular foundation)	8.043
polaris_adme_ fang_solu_1	Polaris	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.575	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	12.303
tdc_caco2_wang	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	0.336	Elastic NetCV	Conventional ML	13.842
tdc_clearance_ hepatocyte_az	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	43.976	TabPFNReg	TabPFN (tabular foundation)	17.126
tdc_clearance_ microsome_az	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	35.888	Tabular MLP	Conventional ML	13.242
tdc_half_life_ obach	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	17.222	TabPFNReg	TabPFN (tabular foundation)	11.498
tdc_ hydrationfreeenergy_ freesolv	TDC	regression	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ rmse	0.556	Elastic NetCV	Conventional ML	143.019
tdc_ld50_zhu	TDC	regression	CFA combinatorial fusion	CFA (Combinatorial Fusion)	test_ rmse	0.834	XGBoost	Conventional ML	1.097
tdc_ lipophilicity_ astrazeneca	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.587	Elastic NetCV	Conventional ML	27.840
tdc_ppbr_az	TDC	regression	Ensemble (stacking / averaging)	Ensemble (Weighted average (inverse train RMSE))	test_ rmse	11.014	ChemML MLP (Py Torch)	Deep tabular NN (ChemML MLP)	28.555

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dataset	suite	task - kind	family	model	analysis_ metric	analysis_metric_value	cv_ selected_ model	cv_ selected_ family	cv_selected_gap_to_best
tdc_solubility_ aqsolddb	TDC	regression	Uni-Mol V1 (3D pretrained)	Uni-Mol V1	test_ rmse	0.989	Extra trees	Conventional ML	5.427
tdc_vdss_ lombardo	TDC	regression	MapLight + GNN	MapLight + GNN (Cat Boost, Strict Parity)	test_ rmse	4.668	TabPFNReg TabPFN	TabPFN (tabular foundation)	16.999

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

stage_order	stage	datasets_evaluated	datasets_improved_vs_previous	improvement_fraction
1	Conventional ML only	45	0	0.000
2	+ Map Light classic fea- tures	45	7	0.156
3	+ neu- ral/ deep back- ends	45	26	0.578
4	+ CFA fusion	45	8	0.178
5	Full pipeline incl. ensem- bles	45	15	0.333

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

Feature family	Datasets selected	Available (sum)	Selected (sum)	% of selected	Enrichment vs uniform
rdkit	41	8,363	1,441	8.55	3.90
erg	40	12,606	1,248	7.40	2.24
avalon	44	44,617	3,921	23.25	1.99
maccs	41	5,980	301	1.78	1.14
fcfp6	44	42,613	1,783	10.57	0.95
maplight	41	30,122	1,197	7.10	0.90
atom_pair	42	39,820	1,393	8.26	0.79
ecfp6	40	43,433	1,454	8.62	0.76
rdk_path	41	45,888	1,394	8.27	0.69
topological_torsion	43	23,736	713	4.23	0.68
layered	42	43,923	1,084	6.43	0.56
morgan	42	40,996	934	5.54	0.52

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

model	datasets _attempted	datasets _valid
Ensemble (OOF Stacking (RidgeCV))	26	0
TabPFNClassifier	10	10
Chemprop v2 (AttentiveFP, ensemble=1)	45	21
LogisticRegression	22	22
SVC	22	22
Voting Classifier (KNN, SVM)	22	22
MapLight CatBoost (Strict Parity)	23	23
ElasticNetCV	23	23
SVR	23	23
Chemprop v2 (AttentiveFP + Selected descriptors, ensemble=1)	45	23
Voting Regressor (KNN, SVM)	23	23
Tabular CNN	23	23
TabPFNRegressor	35	35
MapLight + GNN (CatBoost, Strict Parity)	45	42
AdaBoost	45	45
ChemML MLP (PyTorch)	45	45
Random forest	45	45
Ensemble (Weighted average (inverse train RMSE))	45	45
Extra trees	45	45
HistGradientBoosting	45	45
Ensemble (OOF Stacking (RidgeCV, 5-fold))	45	45
CatBoost	45	45
CFA (Combinatorial Fusion)	45	45
Tabular MLP	45	45
Uni-Mol V1	45	45
XGBoost	45	45